

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE			
	Supplementary Examination – Summer 2022			
	Course: B. Tech.	Branch: E&TC	Semester: III	
	Subject Code & Name: BTETC302 Electronic Devices and Circuits			
	Max Marks: 60	Date:	Duration: 3 Hr.	
	Instructions to the Students: 1. All the questions are compulsory. 2. Use of non-programmable scientific calculators is allowed. 3. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks	
Q. 1	Solve Any Two of the following.		12	
A)	Draw and explain RC coupled CE Amplifier. What are Effects of bypass and coupling capacitors.	(1)	6	
B)	What is Need for biasing BJT also explain Transistor biasing methods.	(2)	6	
C)	Explain BJT: construction, working, characteristics.	(1)	6	
Q.2	Solve Any Two of the following.		12	
A)	Briefly Explain construction and working principle of D-MOSFET	(2)	6	
B)	Describe FET as an Amplifier and its Analysis (CS).	(3)	6	
C)	Compare JFET, MOSFET and BJT w.r.t. to device and Circuit parameter	(2,3)	6	
Q. 3	Solve Any One of the following.		12	
A)	What is Class B push pull and complementary symmetry amplifier with its efficiency and calculate the power output and power dissipation.	(3)	12	
B)	Compare Class A, Class B, Class C, Class AB and Class D amplifiers.	(4)	12	
Q.4	Solve Any Two of the following.		12	
A)	Briefly Explain Principle of Negative feedback in electronic circuits.	(4)	6	
B)	Explain Voltage series, Voltage shunt, Current series, Current shunt types of Negative feedback	(2)	6	
C)	What are the typical transistor circuits effects of Negative feedback on Input and Output impedance, Voltage and Current gains, and Bandwidth?	(3)	6	
Q. 5	Solve Any Two of the following.		12	
A)	Explain about Barkhausen criterion for oscillator also draw RC Phase shift Oscillators	(4)	6	
B)	Describe the Hartley and Colpitts oscillators.	(4)	6	
C)	Explain about adjustable three terminal positive (LM 317) regulators with neat diagram	(4)	6	
	*** End ***			

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